

EDUCATIONAL PROOF-OF-CONCEPT · DEEP LEARNING · TRANSFER LEARNING

ODIR-5K Retinal Fundus Image Classification

Normal vs. Disease Screening — a patient-level, leakage-safe transfer-learning pipeline with clinically meaningful evaluation

Team CGNT · github.com/GokulRaj0110/CGNT_Discoverathon

Dataset: [andrewmvd/ocular-disease-recognition-odir5k](#) (Kaggle)

THE PROBLEM

Screening queues need a fast, honest first pass

Retinal fundus photography is a cheap, non-invasive way to capture signs of diabetic retinopathy, glaucoma, AMD, cataract, hypertension and myopia. But every image still needs a clinician's eyes — and screening volumes outpace review capacity.

What an automated first pass could do

Flag each photo as “normal — clear” or “disease — needs review,” so limited clinician time is prioritized toward the images most likely to need it.

Explicitly educational scope

This is a practice exercise in transfer learning and rigorous evaluation — not a validated diagnostic tool.

Original diagnostic categories → collapsed to binary

Normal	Diabetes
Glaucoma	Cataract
AMD	Hypertension
Myopia	Other

0 = Normal **1** = Disease (any)

Six design commitments, set before writing any code

1

Binary screening: label each fundus image Normal (0) or Disease (1)

2

Transfer learning from an ImageNet-pretrained backbone, given limited labeled medical data

3

Explicit class-imbalance handling in both sampling and loss

4

Patient-level splitting — a patient's two eyes never cross the train/validation boundary

5

Clinically relevant reporting: sensitivity, specificity, F1, confusion matrix, ROC-AUC as primary metric

6

Grad-CAM explainability to sanity-check what the model is actually looking at

DATASET

ODIR-5K: multi-hospital, patient-paired fundus photos

andrewmvd/ocular-disease-recognition-odir5k (Kaggle) — ophthalmologist-labeled, both eyes per patient, multiple hospitals and camera systems (realistic multi-site variability). Binary labels built primarily from full_df.csv, falling back to the patient-level data.xlsx metadata.

~5,000

patients (paired L/R eyes)

8 → 2

diagnostic categories collapsed to binary

7,000

images in the OOF evaluation set

Out-of-fold class support (N = 7,000)

Class	Support
Normal	3,052
Disease	3,948
Total	7,000

Why the split matters

Each patient contributes a left and a right eye. A naive random split can place both in different folds — leaking patient anatomy and imaging conditions between train and validation, and inflating validation performance.

Preprocessing, leakage-checked split, and balancing

1

CLAHE Preprocessing

Contrast-Limited Adaptive Histogram Equalization brings out low-contrast lesions and vessels against the retinal background, plus standard ImageNet normalization to match the pretrained backbone.

2

5-Fold StratifiedGroupKFold

Grouped by patient ID, stratified on the binary label. An explicit leakage check confirms no patient's images ever appear in both train and validation within the same fold.

3

Augmentation

Random crop, horizontal flip, rotation, color jitter, and RandomErasing improve robustness to the dataset's multi-camera, multi-site variability.

4

Class Balancing

A WeightedRandomSampler oversamples the minority class during training — complementing, not replacing, loss-level imbalance handling.

Gradual, group-by-group backbone unfreezing

Instead of unfreezing the whole ResNet50 backbone at once after a brief warm-up, unfreeze one group at a time — output-side first — each with its own discriminative learning rate. This directly replaced an earlier all-at-once approach that overfit badly.



every $EPOCHS_PER_UNFREEZE_GROUP$ epochs · LR decays $\times UNFREEZE_LR_DECAY$ per newly-opened group · stem gets an explicit $FINAL_STAGE_LR$ override

Key hyperparameters

Parameter	Value	Meaning
N_UNFREEZE_GROUPS	4	Backbone chunks staged
EPOCHS_PER_UNFREEZE_GROUP	3	Epochs between unfreeze events
UNFREEZE_LR_DECAY	0.5	Per-group LR decay
FINAL_STAGE_LR	5e-5	Stem's explicit LR
EARLY_STOP_PATIENCE	6	Epochs w/o val-AUC gain

Why this design

Deeper, task-specific layers adapt fast; generic low-level filters change slowly — the schedule most models never bother to try.

RESULTS

Out-of-fold performance across all 5 folds

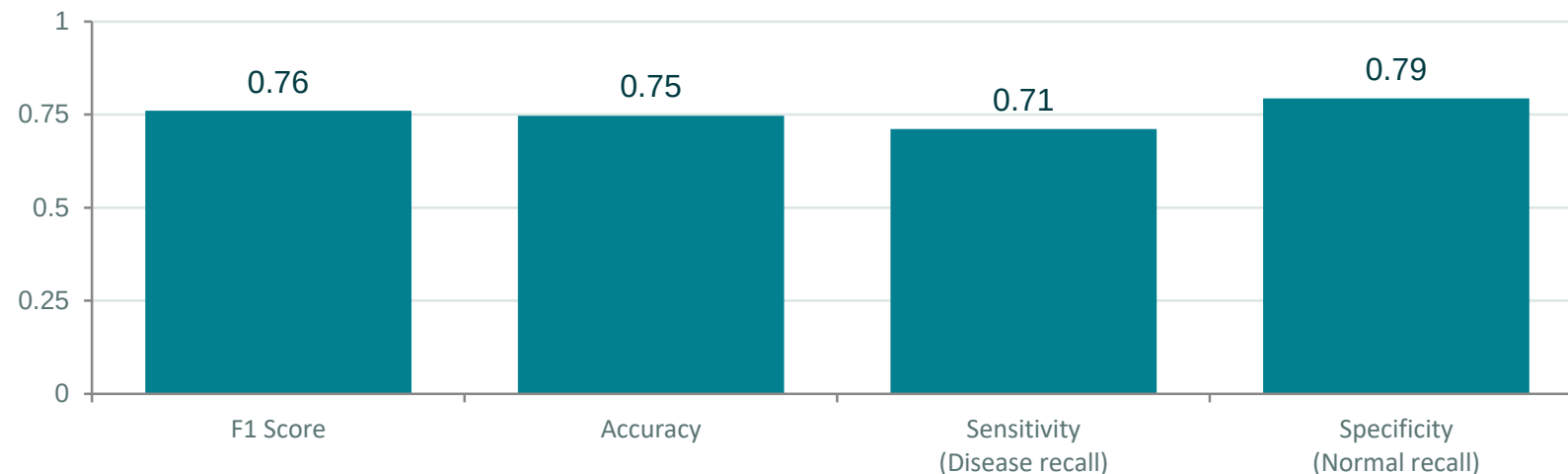
ROC-AUC (primary)

0.822

per-fold mean 0.8247 ± 0.0133

Std across 5 folds is small — a stable, reproducible result, not a lucky split.

Headline Metrics (OOF)



Reading it clinically

At the default 0.5 threshold, the model clears Normal images (0.79 recall) more reliably than it flags Disease images (0.71 recall). For a screening/triage use case — where missing disease is costlier than a false alarm — the operating threshold should be tuned downward before any real triage use, even in a non-clinical prototype.

Where the errors actually land

Predicted →

	Normal	Disease
Actual Normal	2,423 True Negative	629 False Positive
Actual Disease	1,140 False Negative	2,808 True Positive

$t = 0.5 \cdot N = 7,000$

1,140 missed disease images (false negatives) — the more clinically costly error, and the one a lower decision threshold would directly target.

Per-class breakdown (OOF)

Class	Precision	Recall	F1	Support
Normal	0.68	0.79	0.73	3,052
Disease	0.82	0.71	0.76	3,948
Macro avg	0.75	0.75	0.75	7,000

Grad-CAM: a sanity check, not just a picture

Applied to the best fold to confirm attention falls on vessels, optic disc, and macula — not imaging artifacts or camera-specific cues.

Overfitting is reduced, not solved — and we know the next lever

BEFORE — all-at-once unfreeze

Training loss fell toward zero while validation loss climbed through the back half of every fold. Best checkpoint landed early — remaining epochs actively hurt generalization.

AFTER — gradual group unfreeze

Staged unfreezing + discriminative LRs + AUC-based early stopping → smaller, more stable train/val gap (fold AUC std 0.0133). No more runaway validation-loss climb.

Next lever: if val-AUC still drops right when the stem unfreezes, try `N_UNFREEZE_GROUPS = 3` — i.e. never unfreeze the stem, since low-level ImageNet filters are the most overfit-prone on a smaller, domain-shifted dataset.

KEY LIMITATIONS

- No external test set — all metrics are OOF within ODIR-5K's own distribution
- Binary collapse hides which of 8 diseases drives errors
- Sens/spec trade-off is specific to the untuned 0.5 threshold
- Screening aid only — not validated for clinical decision-making

FUTURE WORK (priority order)

- 1 Try `N_UNFREEZE_GROUPS = 3` vs. current 4-group schedule
- 2 Explicitly tune the decision threshold for sensitivity
- 3 Validate on an external, independently sourced dataset
- 4 Break binary back into per-disease evaluation

CONCLUSION

A working, honestly-evaluated screening baseline

A patient-level, leakage-safe ResNet50 pipeline reaches OOF ROC-AUC 0.82 on ODIR-5K, with imbalance handling and a purpose-built gradual-unfreeze training schedule that measurably — though not completely — tames overfitting versus the original all-at-once approach.

INTEGRITY DECLARATION

In the interest of transparency, this project was completed with the assistance of a large language model (LLM) as a collaborative aid during development — including help thinking through pipeline structure, debugging code, discussing design trade-offs such as the unfreezing schedule, and drafting this report and presentation.

The core methodological decision — moving from all-at-once backbone unfreezing to gradual, group-by-group unfreezing with discriminative learning rates — was implemented and tuned directly, based on the overfitting observed in the earlier version. The LLM's role was assistive; the experimentation and judgment behind that decision were not delegated.

Team CGNT